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INNOVATION. AUTOMATION. ANALYTICS

Home Loan Approval Prediction Dataset Overview

GitHub URL : https://github.com/Adityasai1431/Home_Loan_Analysis
LinkedIn URL: https://www.linkedin.com/posts/aditya-kumar-lotti

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Dataset Overview

- The dataset contains 614 rows and 13 columns representing loan application records.
- Columns include: Loan_ID, Gender, Married, Dependents, Education, Self_Employed, ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History, Property_Area, Loan_Status.
- Data types include 8 categorical columns and 5 numerical columns.
- Numerical columns: ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History.
- Categorical columns: Loan_ID, Gender, Married, Dependents, Education, Self_Employed, Property_Area, Loan_Status.

Simple description of each column in your dataset:

- **Loan_ID**: A unique identifier assigned to each loan application.
- **Gender**: Gender of the applicant (Male/Female).
- **Married**: Indicates whether the applicant is married (Yes/No).
- **Dependents**: Number of dependents the applicant has.
- **Education**: Educational background of the applicant (Graduate / Not Graduate).
- **Self_Employed**: Indicates whether the applicant is self-employed (Yes/No).
- **ApplicantIncome**: Monthly income of the primary applicant.
- **CoapplicantIncome**: Monthly income of the co-applicant, if any.
- **LoanAmount**: Amount of loan requested by the applicant (in thousands).
- **Loan_Amount_Term**: Duration of the loan repayment period (in months).
- **Credit_History**: Credit history status of the applicant (1 = good credit history, 0 = poor credit history).
- **Property_Area**: Type of area where the property is located (Urban, Semiurban, Rural).
- **Loan_Status**: Final loan approval status (Y = Approved, N = Not Approved).

Reading Dataset

```
# 1. Load the dataset
loans = pd.read_csv(r"C:\Users\PATEL\Downloads\loan_sanction_train.csv")
loans
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_T
0	LP001002	Male	No	0	Graduate	No	5849	0.0	NaN	3
1	LP001003	Male	Yes	1	Graduate	No	4583	1508.0	128.0	3
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	66.0	3
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	120.0	3
4	LP001008	Male	No	0	Graduate	No	6000	0.0	141.0	3
...	...	...	...	...	...	...	...	...	...	
609	LP002978	Female	No	0	Graduate	No	2900	0.0	71.0	3
610	LP002979	Male	Yes	3+	Graduate	No	4106	0.0	40.0	1
611	LP002983	Male	Yes	1	Graduate	No	8072	240.0	253.0	3
612	LP002984	Male	Yes	2	Graduate	No	7583	0.0	187.0	3
613	LP002990	Female	No	0	Graduate	Yes	4583	0.0	133.0	3

614 rows × 13 columns

Basic Checks

Columns of the Dataset

```
loans.columns
```

```
Index(['Loan_ID', 'Gender', 'Married', 'Dependents', 'Education',
       'Self_Employed', 'ApplicantIncome', 'CoapplicantIncome', 'LoanAmount',
       'Loan_Amount_Term', 'Credit_History', 'Property_Area', 'Loan_Status'],
      dtype='object')
```

Checking Structure of the Dataset

```
# 3. Check structure
loans.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 614 entries, 0 to 613
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Loan_ID            614 non-null    object
 1   Gender             601 non-null    object
 2   Married            611 non-null    object
 3   Dependents         599 non-null    object
 4   Education          614 non-null    object
 5   Self_Employed      582 non-null    object
 6   ApplicantIncome    614 non-null    int64
 7   CoapplicantIncome  614 non-null    float64
 8   LoanAmount         592 non-null    float64
 9   Loan_Amount_Term   600 non-null    float64
 10  Credit_History     564 non-null    float64
 11  Property_Area      614 non-null    object
 12  Loan_Status        614 non-null    object
dtypes: float64(4), int64(1), object(8)
memory usage: 62.5+ KB
```

Checking Summary Statistics of Numerical Columns

```
# 4. Get summary stats
loans.describe()
```

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
count	614.000000	614.000000	592.000000	600.00000	564.000000
mean	5403.459283	1621.245798	146.412162	342.00000	0.842199
std	6109.041673	2926.248369	85.587325	65.12041	0.364878
min	150.000000	0.000000	9.000000	12.00000	0.000000
25%	2877.500000	0.000000	100.000000	360.00000	1.000000
50%	3812.500000	1188.500000	128.000000	360.00000	1.000000
75%	5795.000000	2297.250000	168.000000	360.00000	1.000000
max	81000.000000	41667.000000	700.000000	480.00000	1.000000

Checking Summary Statistics of Categorical Column

```
loans.describe(include="O")
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	Property_Area	Loan_Status
count	614	601	611	599	614	582	614	614
unique	614	2	2	4	2	2	3	2
top	LP001002	Male	Yes	0	Graduate	No	Semiurban	Y
freq	1	489	398	345	480	500	233	422

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Preprocessing Techniques on Dataset Individual Columns

```
loans.isna().sum()
```

```
Loan_ID                0
Gender                13
Married                3
Dependents            15
Education              0
Self_Employed         32
ApplicantIncome        0
CoapplicantIncome      0
LoanAmount            22
Loan_Amount_Term      14
Credit_History        50
Property_Area          0
Loan_Status            0
dtype: int64
```

```
# Check if there are any identical records in the dataset.
loans.duplicated().sum()
```

```
np.int64(0)
```

- 7 Columns have Missing Values
- No duplicates

- Now ,I want to perform some Preprocessing Techniques on columns like:

✓ Handling Missing Values
✓ Handling Inconsistent Data
✓ Data Type Conversions

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Handling Missing Values

```
# Fill missing values of categorical columns with mode
loans['Gender'] = loans['Gender'].fillna(loans['Gender'].mode()[0])
loans['Married'] = loans['Married'].fillna(loans['Married'].mode()[0])
loans['Dependents'] = loans['Dependents'].fillna(loans['Dependents'].mode()[0])
loans['Self_Employed'] = loans['Self_Employed'].fillna(loans['Self_Employed'].mode()[0])
loans.isna().sum()
```

```
Loan_ID               0
Gender                0
Married               0
Dependents            0
Education             0
Self_Employed         0
ApplicantIncome       0
CoapplicantIncome     0
LoanAmount           22
Loan_Amount_Term     14
Credit_History       50
Property_Area         0
Loan_Status           0
dtype: int64
```

- For missing values of the categorical data columns like
- Gender
- Married
- Dependents
- self Employed

- We replaced missing values of all categorical columns with **mode**.

```
# Fill missing values with median
loans['LoanAmount'] = loans['LoanAmount'].fillna(loans['LoanAmount'].median())
loans['LoanAmount'].isna().sum()
```

```
np.int64(0)
```

Column: Loan Amount
We replaced missing values of Loan Amount with **Median**.

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```python
# We can fill missing values with mode
loans['Loan_Amount_Term'] = loans['Loan_Amount_Term'].fillna(loans['Loan_Amount_Term'].mode()[0])
loans['LoanAmount'].isna().sum()
```

```
np.int64(0)
```

Insights: By checking all the unique values and value counts of Loan_Amount_Term column, we replaced missing values of Loan_A
which is the most common value.

```python
# Fill missing values of numerical columns
# Credit_History column
loans['Credit_History'].unique()
```

```
array([ 1.,  0., nan])
```

```python
loans['Credit_History'].value_counts()
```

```
Credit_History
1.0    475
0.0     89
Name: count, dtype: int64
```

```python
# We can fill missing values with mode
loans['Credit_History'] = loans['Credit_History'].fillna(loans['Credit_History'].mode()[0])
loans['Credit_History'].isna().sum()
```

```
np.int64(0)
```

- **<u>Column</u>**: Loan_Amount_Amount
- We replaced missing values of Loan_Amount_Amount with **Mode**.

- **<u>Column</u>**: Credit_History
- We replaced missing values of Credit_History with **Mode**.

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Fixing Inconsistencies

3. Fix Inconsistencies in Categorical Data

```
# Review text data for inconsistency in spelling, case (uppercase/lowercase), or formatting.

for i in loans.select_dtypes('object').columns[1:]:
    print(loans[i].value_counts())
    print('----------------------------')
```

```
Gender
Male      502
Female    112
Name: count, dtype: int64
----------------------------
Married
Yes    401
No     213
Name: count, dtype: int64
----------------------------
Dependents
0     360
1     102
2     101
3+     51
Name: count, dtype: int64
----------------------------
Education
Graduate        480
Not Graduate    134
Name: count, dtype: int64
```

```
Self_Employed
No     532
Yes     82
Name: count, dtype: int64
----------------------------
Property_Area
Semiurban    233
Urban        202
Rural        179
Name: count, dtype: int64
----------------------------
Loan_Status
Y    422
N    192
Name: count, dtype: int64
----------------------------
```

- There are no inconsistencies in the Dataset

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Performing Data Conversions

```
# convert 'Loan_Amount_Term' and 'Credit_History' to categorical data type
loans['Loan_Amount_Term'] = loans['Loan_Amount_Term'].astype('int').astype('category')
loans['Credit_History'] = loans['Credit_History'].astype('int').astype('category')
```

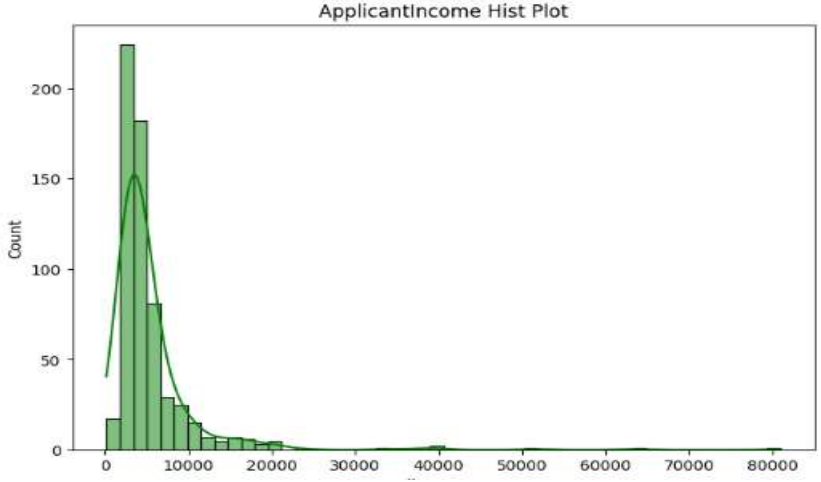
```
# Change categorical columns to category data type
for i in loans.select_dtypes('object').columns[1:]:
    loans[i] = loans[i].astype('category')

loans.dtypes
```

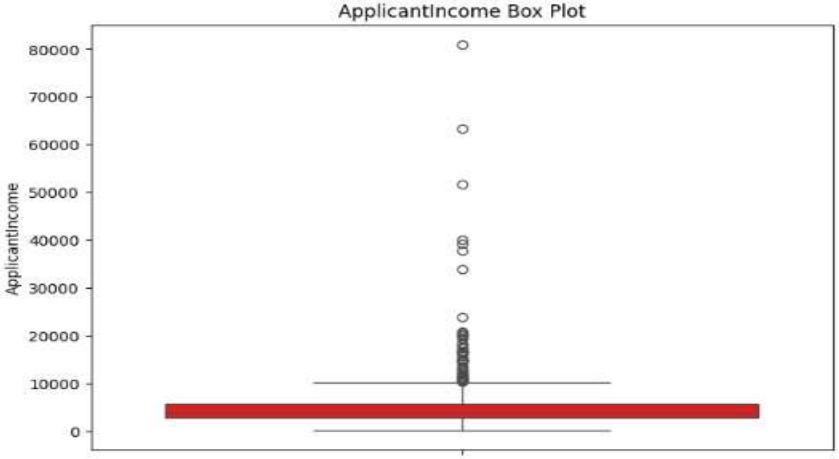
```
Loan_ID                object
Gender               category
Married              category
Dependents           category
Education            category
Self_Employed        category
ApplicantIncome         int64
CoapplicantIncome     float64
LoanAmount            float64
Loan_Amount_Term     category
Credit_History       category
Property_Area        category
Loan_Status          category
dtype: object
```

- We converted all the **object** columns to **category** data type except the **Loan_ID** column and the **Loan_Amount_Term** and **Credit_History** columns to **category** data type.

Data Analysis through Visualizations

➢ Univariate Analysis





- **ApplicantIncome**:
  - The distribution is strongly right-skewed.
  - Most applicant incomes lie between 0 and 5,000.
  - The median ApplicantIncome is approximately 3,800, which lies well within the 0–5,000 range.
  - Several extreme outliers are present, with incomes extending up to 80,000.

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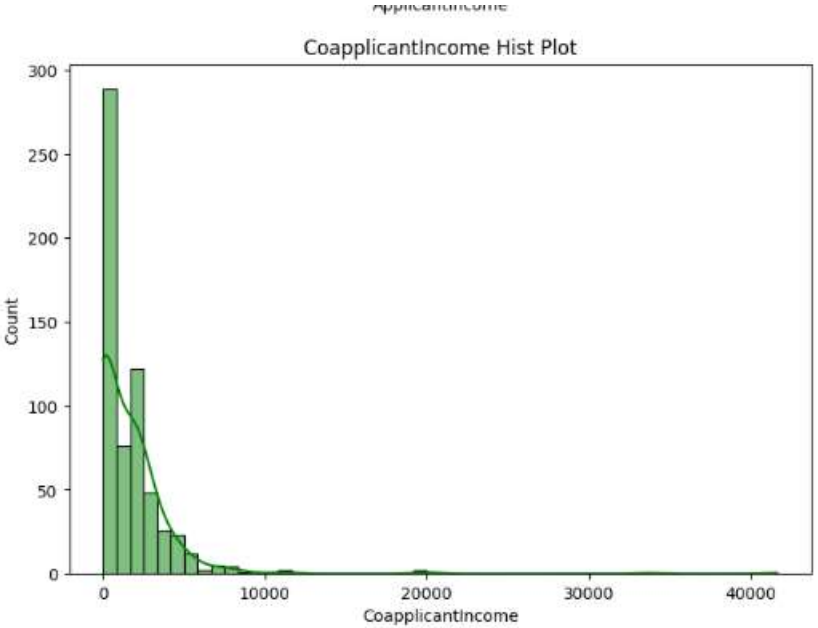

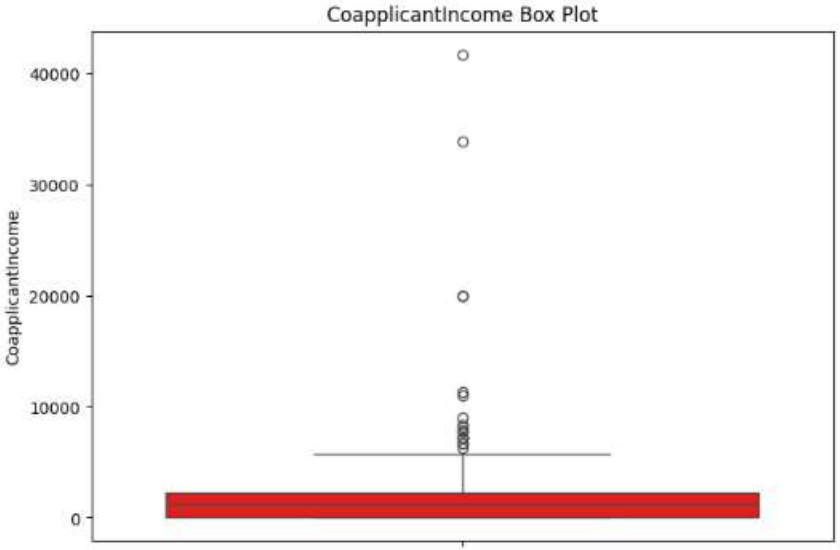

- **CoapplicantIncome**:
  - The distribution is highly right-skewed with a large spike at 0.
  - The median CoapplicantIncome is approximately 1,200.
  - Most values fall between 0 and 4,000, with a long tail extending up to 41,000.
  - Multiple outliers suggest some applicants rely heavily on co-applicant income.

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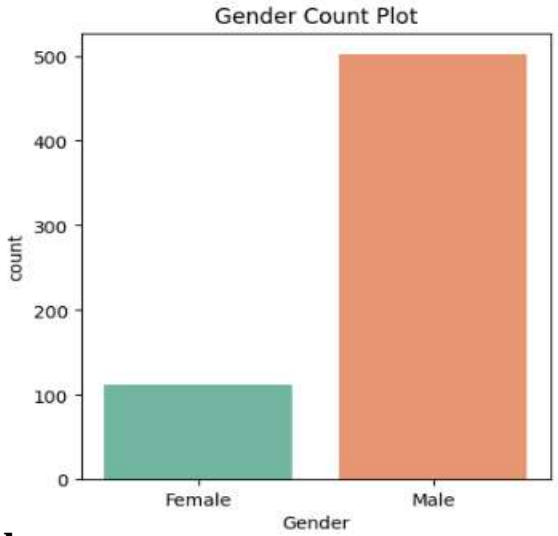
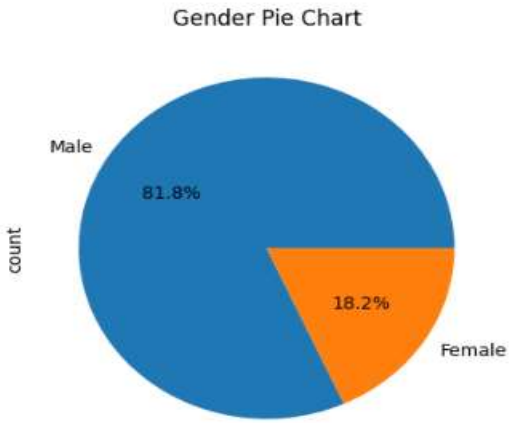
Similarly,

- **LoanAmount**:
  - The distribution is moderately right-skewed.
  - The median LoanAmount is approximately 128.
  - Most loan amounts fall between 100 and 200.
  - Several high-value outliers are present, with loan amounts reaching up to 700.

- **Loan_Amount_Term:**
  - The distribution is highly concentrated, with a strong peak at 360 months, indicating that this is the most common loan tenure.
  - The median Loan_Amount_Term is 360 months, showing that half of the loans are issued for a 30-year duration.
  - Shorter loan terms (such as 12, 36, 60, and 120 months) are present but occur far less frequently.
  - A few outliers exist at very low and very high tenures (up to 480 months), indicating some non-standard loan agreements

Categorical Columns:





- **Gender**
  - The dataset is **male-dominated**, with **81.8% Male** and **18.2% Female** applicants.
  - This imbalance suggests that loan applications are more commonly submitted by males.
  - Gender-based analysis may reflect societal or employment patterns influencing loan access.

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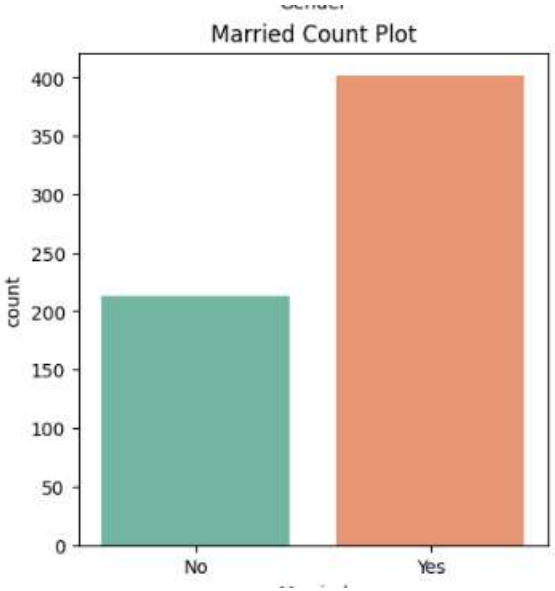
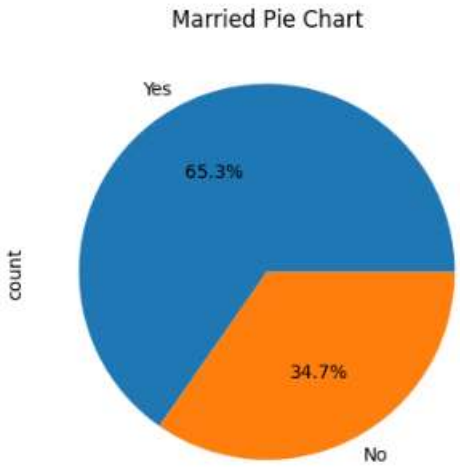


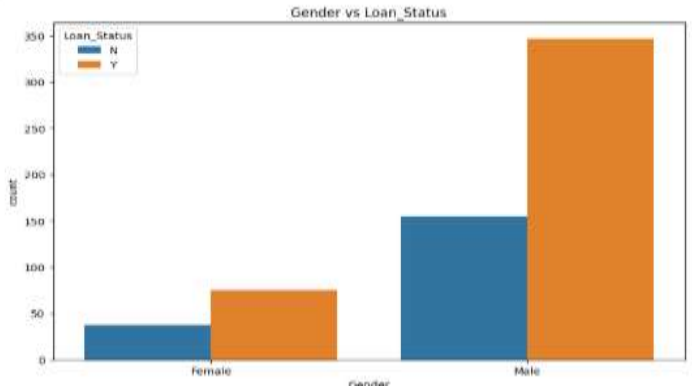


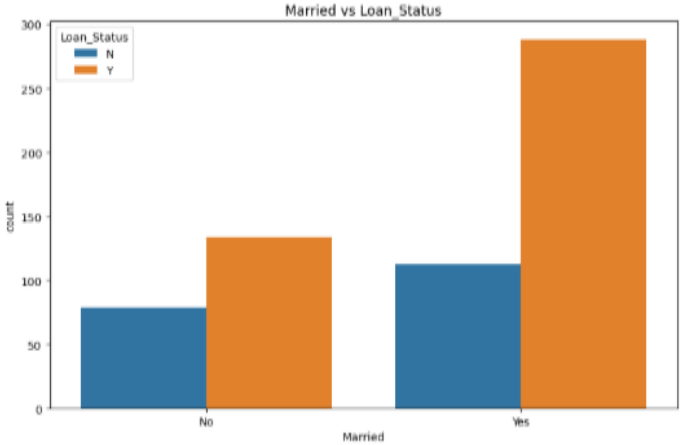
- **Married**
  - A majority of applicants are **Married (65.3%)**, while **34.7% are Not Married**.
  - Married applicants form nearly two-thirds of the dataset, indicating household-based borrowing is common.
  - This may relate to combined income or family financial responsibilities.

- **Education**
  - **Graduates dominate the dataset (78.2%)**, compared to **21.8% Not Graduates**.
  - This suggests that higher education levels are more prevalent among loan applicants.
  - Education may indirectly influence income levels and loan eligibility.
- **Self_Employed**
  - Most applicants are **Not Self-Employed (86.6%)**.
  - Only **~13.4%** are self-employed, indicating salaried individuals form the majority.
  - Lenders may encounter fewer self-employed applicants due to income stability concerns.
- **Property_Area**
  - Applicants are fairly distributed across areas:
  - **Semiurban: 37.9%**
  - **Urban: 32.9%**
  - **Rural: 29.2%**
  - Semiurban areas contribute the highest number of applicants.
  - This suggests growing credit demand in semiurban regions.

- **Loan_Status (Target Variable)**
  - **68.7% loans are Approved (Y)**, while **31.3% are Rejected (N)**.
  - The approval rate is relatively high, indicating favorable lending conditions.
  - This imbalance should be considered during predictive modeling
- **Dependents**
  - **Applicants with 0 dependents dominate (58.6%)**.
  - Applicants with **1 (16.6%)** and **2 (16.4%)** dependents form similar proportions.
  - **3 dependents (8.3%)** is the least common category.
  - Smaller household sizes are more frequent among loan applicants.
- **Credit_History**
  - A strong majority of applicants have **good credit history (1.0) 85.5%**.
  - Only **14.5%** have poor or missing credit history (0.0).
  - Credit history is highly skewed toward positive values, highlighting its importance in loan approval.

Bivariate Analysis-Categorical and Categorical





- **Gender vs Loan Status**
  - Male applicants receive **significantly more loan approvals** compared to female applicants.
  - For both genders, **approved loans (Y)** are higher than rejected loans (N).
  - This suggests a **higher participation and approval rate among males**, possibly due to income or employment differences.
- **Married vs Loan Status**
  - **Married applicants have a much higher loan approval rate** than unmarried applicants.
  - While unmarried applicants also receive approvals, the proportion is lower.
  - This indicates that **marital status may positively influence loan approval**, possibly due to dual income or financial stability.

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Similarly,

- **Dependents**

Applicants with fewer dependents have higher loan approval chances.

- **Education**

Graduates receive more loan approvals than non-graduates.

- **Self-Employed**

Salaried applicants are approved more often than self-employed applicants.

- **Credit History**

Good credit history is the strongest factor for loan approval.

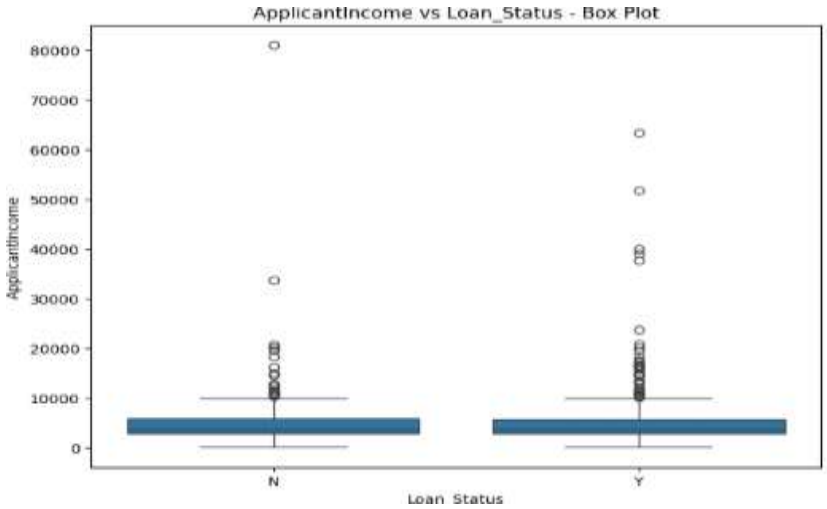
- **Loan Amount Term**

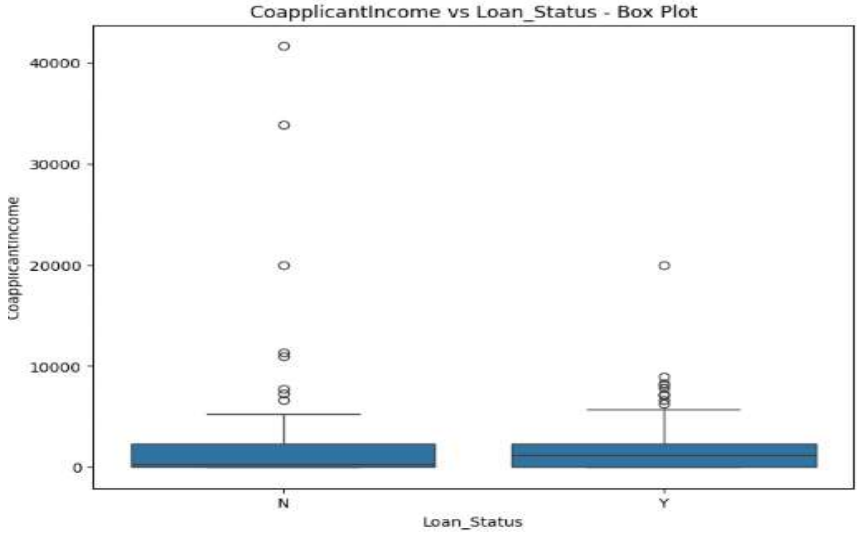
Loan term has little to no impact on approval decisions.

- **Property Area**

Semiurban and Urban applicants have higher approval rates than Rural.

Bivariate Analysis-Categorical and Numerical





◆ **Applicant Income**

- Median income is similar for approved and rejected loans.
- High income alone does not guarantee approval.

◆ **Coapplicant Income**

- Approved loans show slightly higher coapplicant income.
- Additional income support improves approval chances.

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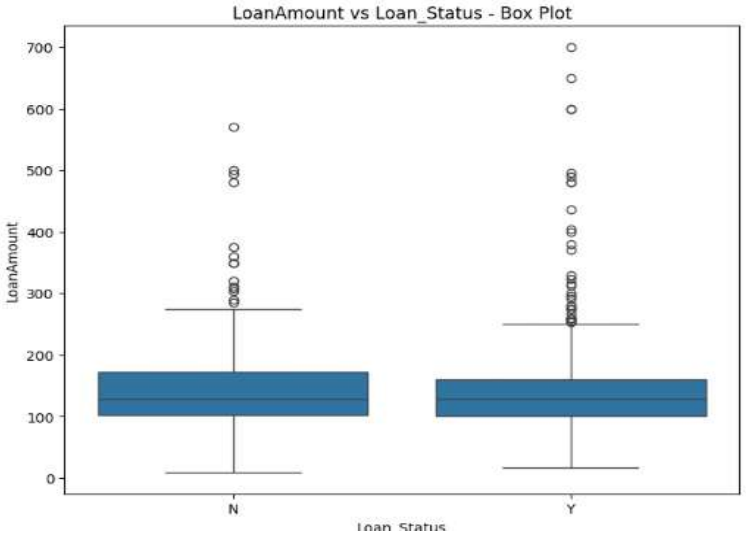

- **LoanAmount & Loan Status**
  - Approved loans tend to have a slightly lower average loan amount than rejected loans.
  - Larger loan amounts appear more frequently among rejected applications.
  - This suggests that higher loan demand may reduce approval chances.

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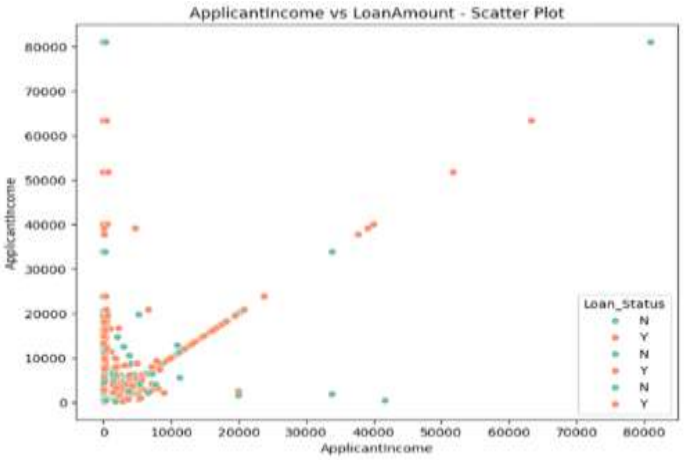
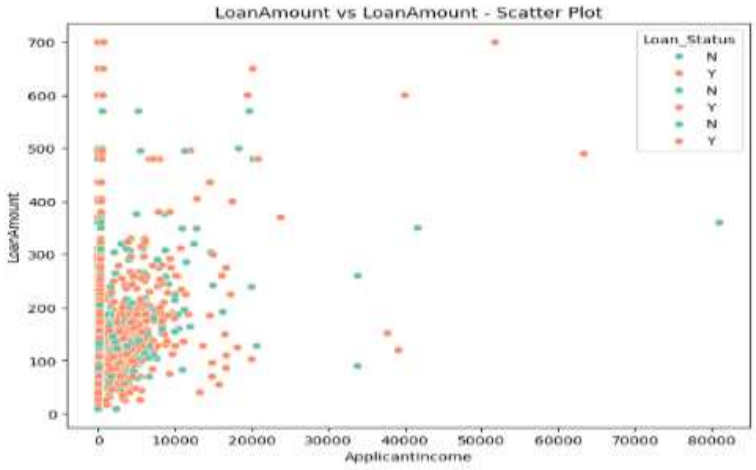
Bivariate Analysis-Numerical and Numerical





➢ Applicant Income

- Loan approvals occur across all income levels.
- High income alone does not ensure approval.

➢ Coapplicant Income

- Presence of coapplicant income supports approval chances.
- Zero or low coapplicant income appears more in rejections.

➢ Coapplicant Income

- Approved loans are mostly concentrated around lower to mid-range loan amounts.
- Higher loan amounts show more scattered rejections, suggesting increased risk.

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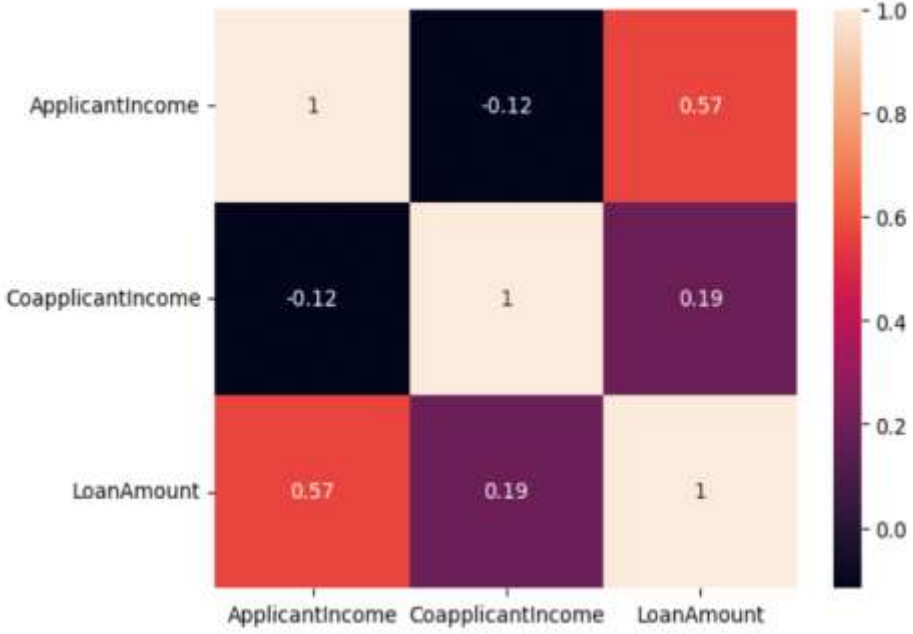
Multivariate Analysis

```
sns.heatmap(loans[num_cols].corr(), annot=True)
```

```
<Axes: >
```



➢ Correlation Insight

- ApplicantIncome and LoanAmount show a moderate positive correlation (0.57), meaning higher income often leads to higher loan requests.
- CoapplicantIncome and LoanAmount show a moderate positive correlation (0.50), indicating that coapplicant income has a positive impact on loan approval.
- Overall, numerical variables influence loan approval indirectly, with credit history and categorical factors likely playing a stronger role.

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Final Summary and Conclusion

This project conducted a comprehensive Exploratory Data Analysis (EDA) on the **Home Loan Approval** dataset to understand the key factors influencing loan approval decisions and applicant behavior.

Key Analytical Steps:

- **Data preparation** ensured clean, consistent, and duplicate-free records by handling missing values and correcting data types.
- **Univariate analysis** examined the distribution of individual variables such as gender, marital status, education, dependents, applicant income, loan amount, credit history, property area, and loan status.
- **Bivariate analysis (visual and non-visual)** explored relationships between:
  - Credit history and loan approval status
  - Applicant income and loan amount
  - Education, employment type, and loan approval
  - Property area and approval trends
- **Categorical analysis** focused on understanding approval patterns across demographic and socioeconomic groups.
- **Numerical analysis** evaluated correlations among income, loan amount, and loan term to identify financial behavior patterns.

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Key Insights

- **Credit history** is the strongest predictor of loan approval, with applicants having a good credit record being approved in most cases.
- **Applicant income** influences the **loan amount requested**, but does not strongly determine approval by itself.
- **Graduates and salaried** (non-self-employed) applicants show higher approval rates, indicating preference for financial stability.
- **Married applicants** and those with **fewer dependents** tend to have better approval outcomes.
- **Semiurban** and **urban property** areas exhibit higher loan approval rates compared to rural areas.
- A higher **cost (loan amount)** does not necessarily reduce approval when credit history is strong.

Final Outcome

- Applicants with **strong credit history, stable employment, and balanced financial profiles** emerge as the most likely candidates for loan approval.
- The analysis shows that lenders prioritize **risk mitigation factors** over income alone, emphasizing long-term repayment reliability.

Conclusion

- The Loan Approval dataset demonstrates that loan eligibility is **multidimensional**, driven primarily by **creditworthiness**, supported by education, employment stability, and demographic factors.
- Effective lending decisions should balance **financial indicators** and **applicant stability** rather than relying solely on income levels.
- This EDA provides a solid foundation for **predictive modeling, credit risk assessment, and policy optimization** in future work.

Thank You

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